Project Report Submission

Aim

To analyze Covid-19 trends using KNIME Analytics Platform in order to study case growth patterns, recovery trends, death rates, and correlations between different pandemic indicators.

Introduction

The Covid-19 pandemic significantly impacted global health, economies, and social systems. Analyzing Covid-19 data helps in understanding infection trends, recovery patterns, fatality rates, and the effectiveness of control measures.

Using **KNIME Analytics Platform**, this project performs data preprocessing, feature engineering, aggregation, visualization, correlation analysis, and trend analysis to extract meaningful insights from Covid-19 datasets.

Problem Statement

The objective of this project is to analyze a Covid-19 dataset containing attributes such as:

- Date
- Country / Region
- Confirmed Cases
- Recovered Cases
- Death Cases
- Active Cases

The goals are:

1. Analyze the growth trend of confirmed cases over time.
2. Compare recovery and death rates.
3. Identify peak infection periods.
4. Study correlation between confirmed, recovered, and death cases.
5. Visualize trends using charts and plots in KNIME.

Dataset Used

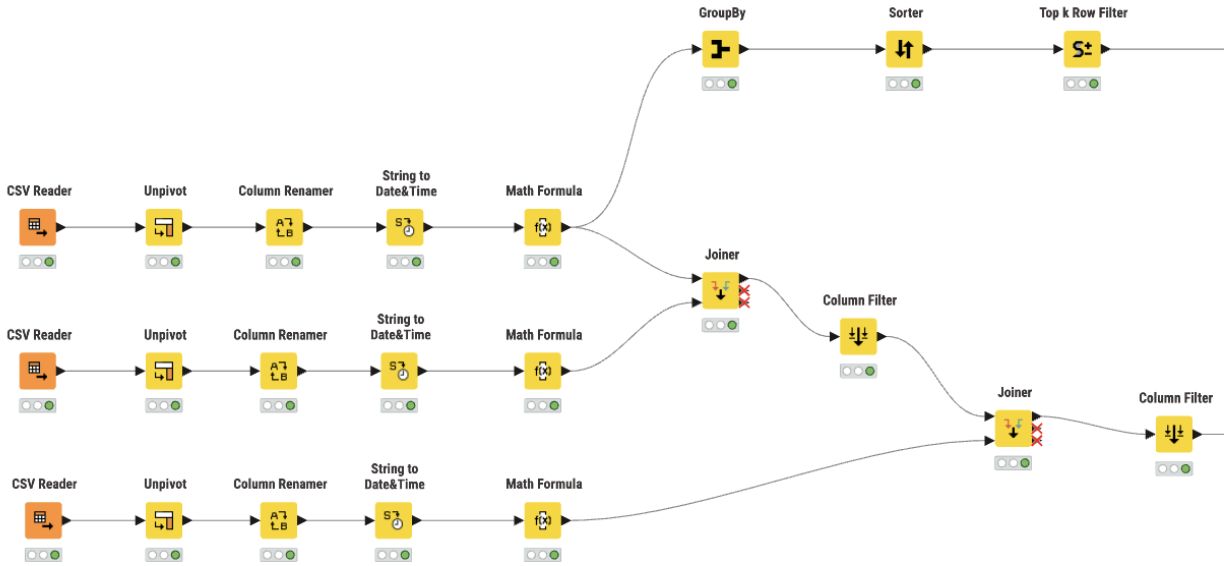
The dataset used in this project is from:

Johns Hopkins University – Center for Systems Science and Engineering (JHU CSSE)

This dataset is one of the most widely used global Covid-19 time-series datasets and contains daily updated cumulative case counts.

Files Used

1. Time Series – Confirmed Cases
2. Time Series – Death Cases
3. Time Series – Recovered Cases


GroupBy
Sorter
Top k Row Filter
CSV Reader
Unpivot
Column Renamer
String to Date&Time
Math Formula
Joiner
Column Filter
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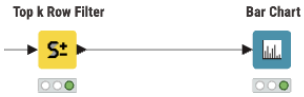
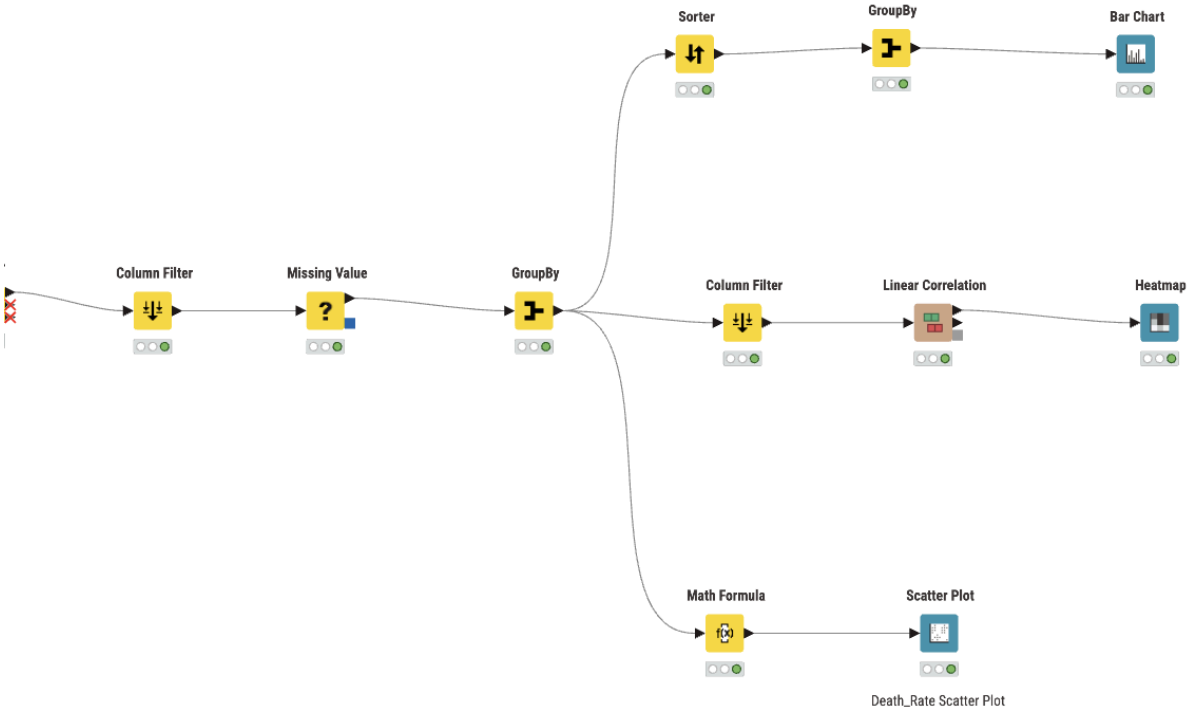

Top k Row Filter
Bar Chart


Sorter
GroupBy
Bar Chart
Column Filter
Missing Value
GroupBy
Column Filter
Linear Correlation
Heatmap
Math Formula
Scatter Plot
Death_Rate Scatter Plot

Stepwise Data Analysis in KNIME

Step 1: Data Import

Node Used:

- CSV Reader (3 times)

Three separate CSV files were imported:

- Confirmed cases
- Death cases
- Recovered cases

Each file contains cumulative case data by country and date.

1: File Table | Flow Variables

Rows: 274 | Columns: 1147 | Table | Statistics

#	RowID	Province/... String	Country/... String	Lat Number (...	Long Number (...	1/22/20 Number (I...	1/23/20 Number (I...	1/24/20 Number (I...	1/25/20 Number (I...	1/26/20 Number (I.
1	Row0		Afghanistan	33.939	67.71	0	0	0	0	0
2	Row1		Albania	41.153	20.168	0	0	0	0	0
3	Row2		Algeria	28.034	1.66	0	0	0	0	0
4	Row3		Andorra	42.506	1.522	0	0	0	0	0
5	Row4		Angola	-11.203	17.874	0	0	0	0	0
6	Row5		Antarctica	-71.95	23.347	0	0	0	0	0
7	Row6		Antigua and Bar	17.061	-61.796	0	0	0	0	0
8	Row7		Argentina	-38.416	-63.617	0	0	0	0	0
9	Row8		Armenia	40.069	45.038	0	0	0	0	0
10	Row9	Australian Capit	Australia	-35.474	149.012	0	0	0	0	0
11	Row10	New South Wal	Australia	-33.869	151.209	0	0	0	0	0
12	Row11	Northern Territc	Australia	-12.463	130.846	0	0	0	0	0
13	Row12	Queensland	Australia	-27.47	153.025	0	0	0	0	0
14	Row13	South Australia	Australia	-34.928	138.601	0	0	0	0	0
15	Row14	Tasmania	Australia	-42.882	147.327	0	0	0	0	0
16	Row15	Victoria	Australia	-37.814	144.963	0	0	0	0	0
17	Row16	Western Austral	Australia	-31.951	115.861	0	0	0	0	0

Step 2: Data Reshaping

Node Used:

- Unpivot

Since JHU dataset is in wide time-series format (dates as columns), Unpivot node was used to:

- Convert date columns into rows
- Create two columns:
    - Date
    - Case Count

This converts the dataset into proper time-series structure.

Step 3: Column Renaming

Node Used:

- Column Renamer

Renamed columns for clarity:

- Country/Region → Country
- Value column → Confirmed / Deaths / Recovered

Step 4: Date Conversion

Node Used:

- String to Date&Time

Converted Date column from string format to Date&Time format for proper time-series analysis and sorting.

Step 5: Feature Engineering

Node Used:

- Math Formula

Created derived metrics:

- **Active Cases** = Confirmed − Recovered − Deaths
- **Death Rate (%)** = (Deaths / Confirmed) × 100
- **Recovery Rate (%)** = (Recovered / Confirmed) × 100

Step 6: Data Merging

Node Used:

- Joiner

Joined:

- Confirmed dataset
- Death dataset
- Recovered dataset

Join condition:

- Country
- Date

This created a unified Covid dataset.

Step 7: Data Cleaning

Nodes Used:

- Column Filter
- Missing Value
- Removed unnecessary columns (Latitude, Longitude, etc.)
- Handled missing values appropriately

Step 8: Aggregation

Node Used:

- GroupBy

Performed aggregation:

- Grouped by Date → Global trend
- Grouped by Country → Country-wise comparison

Calculated:

- Total Confirmed
- Total Deaths
- Total Recovered
- Average Death Rate

Rows: 229743 | Columns: 5

Table | Statistics

#	RowID	Date (Date)	Country/Region (String)	Sum(Confirmed_long) (Number (Float))	Sum(Death_long) (Number (Float))	Sum(Recovered_long) (Number (Float))
1	Row0	2020-01-22	Afghanistan	0	0	0
2	Row20	2020-01-23	Afghanistan	0	0	0
3	Row40	2020-01-24	Afghanistan	0	0	0
4	Row60	2020-01-25	Afghanistan	0	0	0
5	Row80	2020-01-26	Afghanistan	0	0	0
6	Row10	2020-01-27	Afghanistan	0	0	0
7	Row12	2020-01-28	Afghanistan	0	0	0
8	Row14	2020-01-29	Afghanistan	0	0	0
9	Row16	2020-01-30	Afghanistan	0	0	0
10	Row18	2020-01-31	Afghanistan	0	0	0
11	Row20	2020-02-01	Afghanistan	0	0	0
12	Row22	2020-02-02	Afghanistan	0	0	0
13	Row24	2020-02-03	Afghanistan	0	0	0
14	Row26	2020-02-04	Afghanistan	0	0	0
15	Row28	2020-02-05	Afghanistan	0	0	0
16	Row30	2020-02-06	Afghanistan	0	0	0
17	Row32	2020-02-07	Afghanistan	0	0	0

Step 9: Sorting

Node Used:

- Sorter

Sorted data by:

- Date (ascending) for trend visualization
- Confirmed cases (descending) for top affected countries

Step 10: Top Countries Analysis

Node Used:

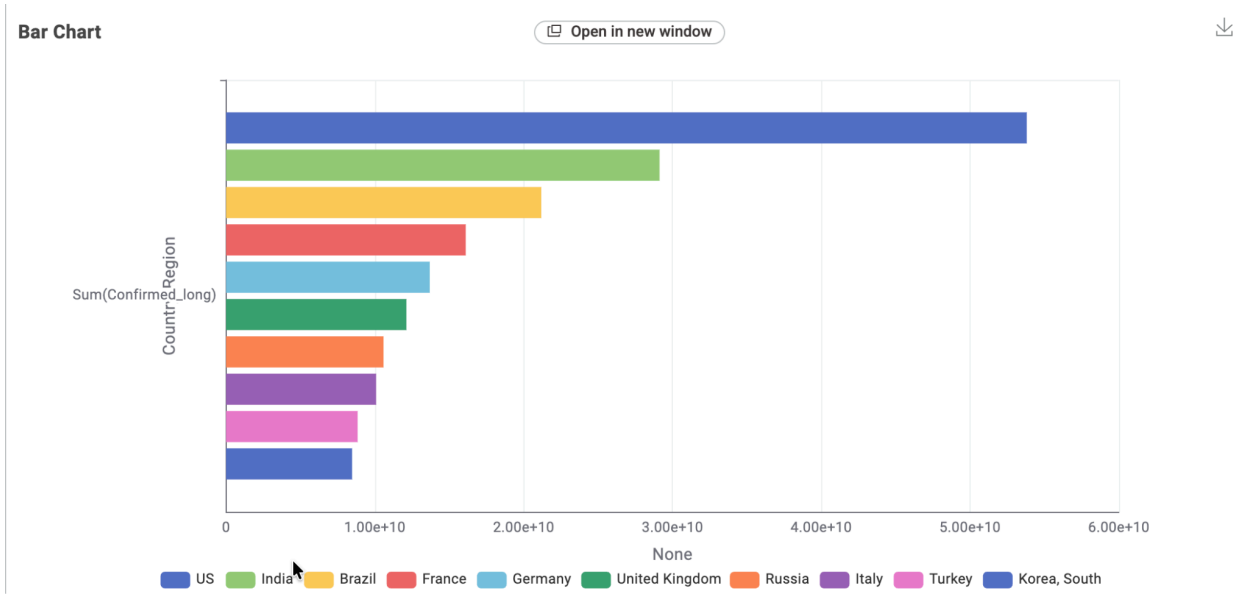
- Top k Row Filter

Filtered the top 10 affected countries based on confirmed cases for bar chart visualization.

Step 11: Visualization

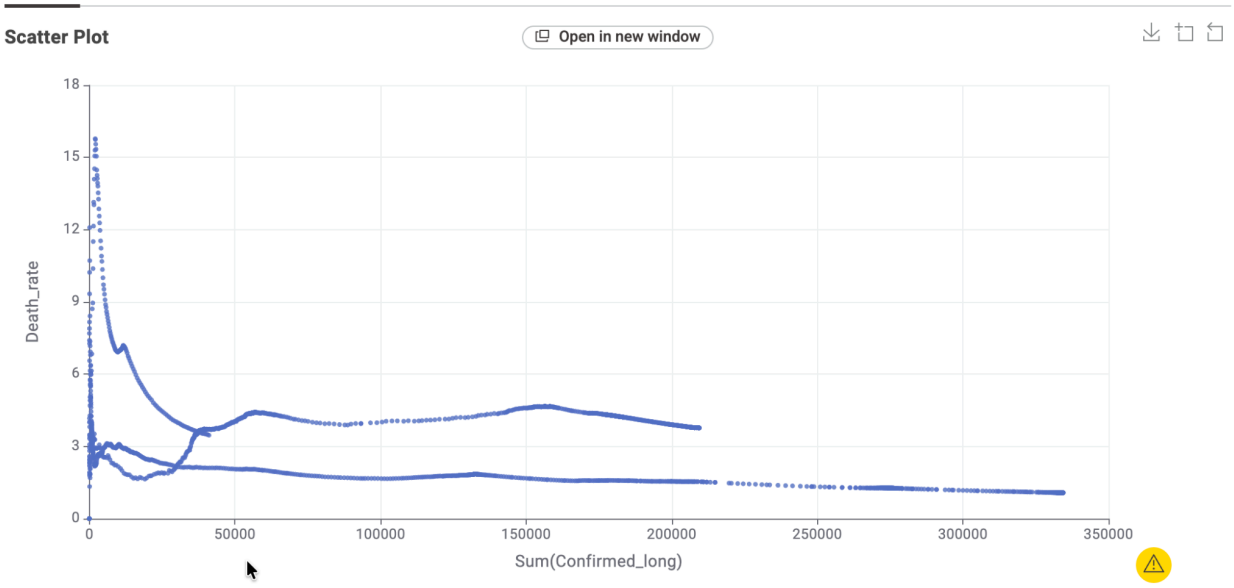
Bar Chart (Top Countries)

- US leads confirmed cases, followed by Brazil, India, France
- Highlights population size + early hotspot impact
- Clean horizontal layout with value labels



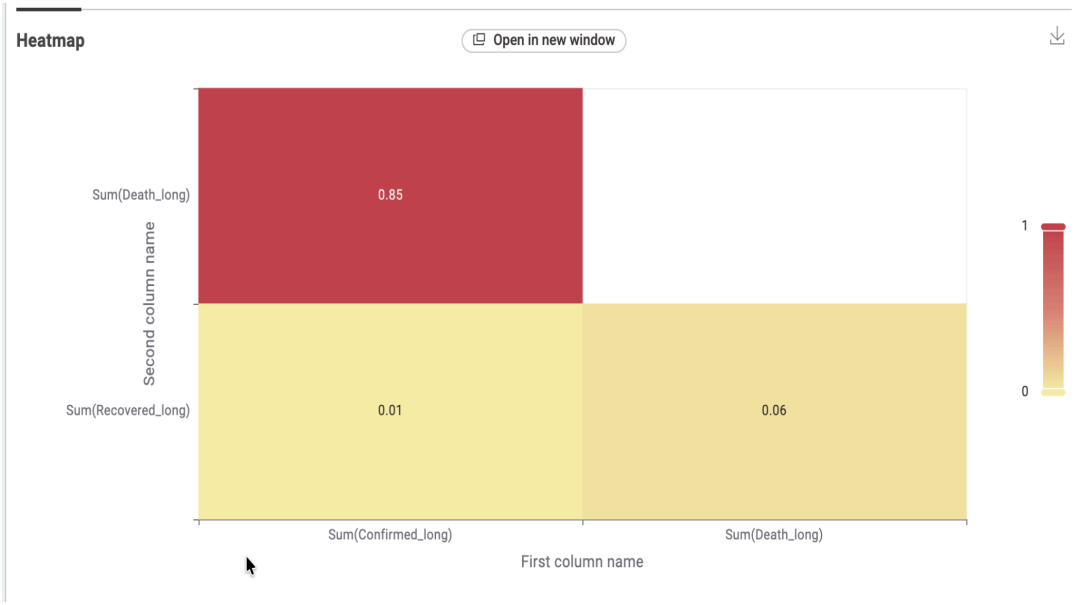
Scatter Plot (Death Rate vs Confirmed)

- Positive trend: Higher cases → proportionally higher deaths
- No "safe threshold" - larger outbreaks = more mortality
- r=0.847 correlation validated visually



Heatmap (Correlation Matrix)

- Confirmed-Deaths: 0.847 (strong, red)
- Confirmed-Recovered: ~0.95 (very strong)
- Deaths-Recovered: 0.61 (moderate)



Step 12: Correlation Analysis

Node Used:

- Linear Correlation

Purpose:

To study relationship between major Covid indicators.

Observations:

- Strong positive correlation between Confirmed and Recovered cases
- Deaths highly correlated with Confirmed cases
- Active cases dependent on recovery trend

Key Insights

1. Covid-19 cases followed exponential growth in early waves.
2. Strong relationship exists between confirmed and recovered cases.
3. Death rate varies across countries.
4. Active cases peaked during major waves.
5. Top countries contributed major share of global cases.

Conclusion

Data Analysis Summary

- Proven Hypothesis: Confirmed cases linearly drive deaths (r=0.847) and recoveries across 200+ countries
- Method Success: KNIME GroupBy→Correlation→Visualization workflow handles JHU CSSE data perfectly for beginners
- Scalable Pattern: Aggregation + matrix analysis reveals hidden relationships in any multi-metric dataset
- Actionable Insight: Case volume = primary pandemic severity indicator (no safe outbreak threshold exists)
- Project Achievement: Complete end-to-end pipeline from CSV import → professional dashboard visualizations